

File Handling (Algo + C)

Statements with solutions in pseudo-algorithm and C

Exercise 1 — Create and write to a text file

Statement. Create `out.txt` and write `Hello` into it.

Exercise 2 — Append lines from user input

Statement. Open `journal.txt` in append mode and copy input lines until an empty one is entered.

Exercise 3 — Copy a file (binary)

Statement. Copy `src` to `dst` byte by byte (buffer).

Exercise 4 — Count lines, words, characters (wc-style)

Statement. Display `#lines`, `#words`, `#characters` of a text file.

Exercise 5 — Count occurrences of a pattern (text)

Statement. Count all occurrences (including overlapping) of a string `pattern`.

Exercise 6 — Display lines containing a pattern (mini-grep)

Statement. Print each line that contains `pattern`.

Exercise 7 — Replace a word and save to a new file

Statement. Replace `src` → `dst` all occurrences of the word.

Exercise 8 — Merge two sorted integer files

Statement. Two files (one integer per line). Produce the sorted merge.

Exercise 9 — Read a simple CSV and compute average and best

Statement. File `name;score`. Compute the average and display the best.

Exercise 10 — Write then read an array of integers in binary

Statement. Input n integers, `fwrite` to `data.bin`, read back and sum.

Exercise 11 — Save/load records (struct) in binary

Statement. `struct Etu{char name[32]; double score;}` write then read.

Exercise 12 — Split a file into chunks of N lines

Statement. Write `part_001.txt`, `part_002.txt`, ... each with N lines.

Exercise 13 — Sort and deduplicate lines (`sort + uniq`)

Statement. Read all lines, sort, then write unique lines.

Exercise 14 — Compare two files (simple binary diff)

Statement. Indicate if they are identical; if not, give the first differing position.

Exercise 15 — Display the last N lines (`tail`)

Statement. Display the last N lines of a file.

Exercise 16 — Convert CRLF → LF

Statement. Replace `\r\n` with `\n`.

Exercise 17 — Simple checksum (djb2) of a file

Statement. Compute a 32-bit hash (djb2) of the binary stream.

Exercise 18 — ASCII frequency histogram

Statement. Count occurrences of each byte (0..255) and display those with non-zero counts.

Exercise 19 — Reverse the contents of a file

Statement. Produce `out` containing the bytes of `in` in reverse order.

Exercise 20 — Prefix a file with a header (create a new file)

Statement. Write `out` = `[header]` followed by the contents of `in`.
